Modeling languages: alpha and omega

João Pedro PEDROSO|DIONÍSIO

Faculty of Sciences, University of Porto

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Outline

1 The "Alpha" of Optimization

The Pre-History: Joseph Fourier (1826)

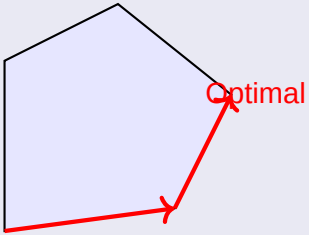
Before Simplex, Fourier devised a method for solving systems of linear inequalities:

- Method: Fourier–Motzkin Elimination.
- Goal: Determining the feasibility of undetermined systems.
- Insight: Optimization (min-max) is equivalent to solving inequalities.

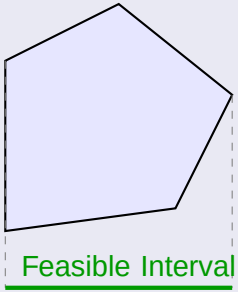
- Fourier (1826): Project the n -dimensional feasible region onto lower dimensions.
  - Problem: Inequality count grows doubly exponentially 2^{2^n} (2
- Dantzig (1947): Move between vertices of the n -dimensional region.
  - Advantage: Usually requires only m to 3m iterations (where m is the number of constrai

Dantzig noted that had he known FME he might have been discouraged by the perceived

Visualizing the Two Approaches



Simplex: Vertex Walk



Fourier: Projection

Why Fourier's Method "Explodes"

- Simplex: Complexity depends on the number of vertices (usually e□cient).
- Fourier-Motzkin: Each variable elimination creates a new set of inequalities.
  - If you have m inequalities, eliminating one variable can result in up to $(m^2/4)$ new ones.
  - The Omega of manual methods: Impossible for Stigler's 77 variables, but theoretically vita
    for proving the Farkas Lemma.

The "Alpha" of Optimization: The Stigler Diet (1947)

The Problem

George Stigler (1945) sought the minimum cost to feed a 154lb man.

- Goal: Minimize daily cost of food.
- Scale: 77 candidate foods, 9 nutritional constraints.

The Mathematical Formulation

- Minimize $Z = \sum_{j=1}^{77} c_j x_j$
- Subject to: $\sum_{j=1}^{77} a_{ij} x_j \geq b_i \quad \forall i \in \{1, \ldots, 9\}, \qquad x_j \geq 0 \quad \forall j \in \{1, \ldots, 77\}$
- Where:
  - c_j: Cost of food j
  - a_{ij}: Amount of nutrient i in food j
  - b_i: RDA for nutrient i (Calories, Protein, Calcium, Iron, Vit A, B1, B2, Niacin, Vit C)

The Human "Computer": Jack Laderman's Feat

In 1947, the "Simplex Method" was brand new (Dantzig). There was no "software."

- Team: 9 human clerks at the National Bureau of Standards.
- Hardware: Hand-operated desk calculators
- E□ort: 120 man-days of computation.

- Stigler's heuristic guess: $39.93/year (1939 prices).
- Laderman's Simplex solution: $39.69/year.
- Fun Fact: The "optimal" diet consisted mostly of wheat flour, evaporated milk, cabbage, spinach, and navy beans. Stigler called it the "human dog biscuit."

The First Machine Implementation: SEAC & IBM 701

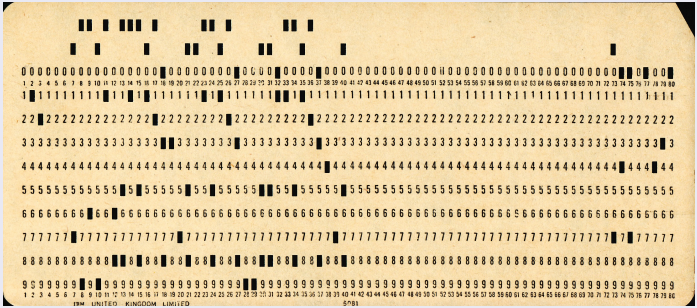
- Machine: Standards Eastern Automatic Computer (NBS).
- Implementation: First general-purpose Simplex code for a stored-program computer.
- Language: Machine Code / Binary (no high-level languages like Fortran existed yet).
- Scale: Solved a 48x71 Air Force model in minutes.

- Developer: William Orchard-Hays at RAND.
- Language: Assembly (highly optimized for the 701's architecture).
- Impact: Could handle ~100 constraints. This code (and the later RSLP1 for IBM 704 moved LP from research to the oil industry (refining/blending).
- The "Omega" Seed: This was the start of Matrix Generators, the tedious precursors t modeling languages we use today.

Before Modeling Languages: Physical Data Entry

In the "Alpha" phase, there was no model.lp or model.py. Data was physical.

- Storage: IBM 80-column punched cards.
- Method: One non-zero coe□cient per card.
- Hardware: IBM 026 Keypunch & 056 Verifier.



Before Modeling Languages: Workflow

1. Mathematical Model: Written on paper.
2. Matrix Tabulation: Coe□cients mapped to row/column indices.
3. Keypunching: Manual transcription to cards.
4. Verification: Mechanical "double-checking" of holes.

Technical Pitfalls

- A typo in one card meant the Simplex algorithm would diverge or find a wrong solution
- There was no "Algebraic" check—only a mechanical one.
- Transition: This pain directly to the creation of Matrix Generators in the 1960s.

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